

WW2 WINGS



NORMANDY · 1944

*A pilot's career in three mini-games.
A tribute to Cinemaware's Wings (1990).*

A career, one sortie at a time

You are a replacement pilot posted to the 406th Fighter Squadron in the first days of June 1944. You fly the P-47 Thunderbolt, and the invasion of France is about to begin.

WW2 Wings follows the shape of Cinemaware's 1990 Amiga classic *Wings*, moved from the Great War to the Second. The story is told through your pilot's diary. Between entries you fly, and every sortie is one of three different mini-games.

- **Air combat** in full 3D, seen from behind your aircraft.
- **Bombing** from directly overhead.
- **Ground attack** at tree-top height, seen from an isometric camera.

Around the flying sits the career: a pilot you create, aptitudes that grow with experience, promotions and medals, and the real chance that he does not come home. When that happens a replacement takes his bunk and the war goes on.



The main menu. The flight of Thunderbolts behind it is rendered live.

3

MINI-GAMES

6

MISSIONS

4

PILOT APTITUDES

5

DECORATIONS

0

ART OR SOUND FILES

Air combat



A Bf 109 goes down trailing smoke through the cloud tops.



Two bandits bracketed ahead. Distant aircraft are drawn larger than life so they can be seen and fought.

Dogfights against Bf 109s and Fw 190s, flown from a chase camera with your wingmen beside you.

- Roll toward the enemy, then pull. Banked wings carry the nose around, and the aircraft eases back to level when you let go of the stick.
- A **lead marker** shows where to shoot; brackets, range and edge arrows keep track of every bandit.
- A radar disc shows who is around you and whether they are above or below.
- Damaged aircraft trail smoke, handle worse and finally go down burning.
- Ammunition is limited. Run dry and you break off for home.

IN THE ORIGINAL

Wings showed its dogfights from behind the pilot's head. The idea is kept; the view is pulled back to a modern chase camera.

Bombing



A stick of bombs walks through the rail yard at Amiens.



A goods train caught on the main line on the way in.

A run over the target from directly above, with a limited bomb load and flak coming up to meet you.

- Bombs take time to fall. A **ring on the ground** shows where one released now will land.
- Rolling stock, warehouses and fuel dumps burn, and fuel sets off whatever stands next to it.
- Flak is aimed at where you are going to be. Keep moving.
- Miss the objective and you can come around for one more pass.
- Two missions: the marshalling yard at Amiens and the bridge over the Orne.

IN THE ORIGINAL

The top-down bombing run is the closest of the three to the 1990 game. The impact marker and the second pass are additions.

Ground attack



Evreux airfield at dusk. A parked fighter goes up beside the runway.



A flak position beside the Caen road, about to be hit.

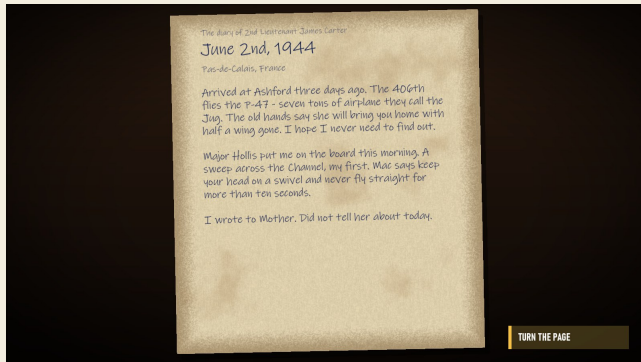
Strafing at tree-top height along a road or an airfield, seen from an isometric camera.

- Slide left and right, climb and dive. Your rounds strike ahead of you, and **the lower you fly the tighter the pattern.**
- Low is also where the poplars, barns and church towers are. Hit one and you may not get a second chance.
- Your shadow on the ground is the best guide to your height.
- Trucks burn easily; tanks shrug off most of what you throw at them.
- Two missions: the Caen road on D-Day and the airfield at Evreux.

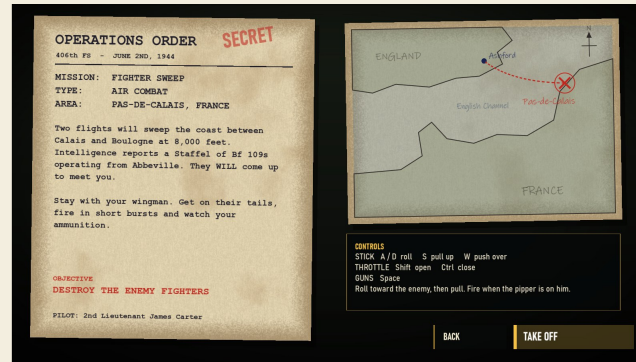
IN THE ORIGINAL

Wings flew its strafing runs diagonally across an isometric landscape, with height as the second axis. That is unchanged.

The career



The diary sets the scene before each mission.



The briefing: orders, route map and controls.



The combat report: results, promotion, medals.

Your pilot. You name him, say where he is from and spend points on four aptitudes. A successful sortie raises the aptitude it exercised.

Flying	How tightly the aircraft turns and how quickly it answers.
Shooting	Tighter bursts and harder hits.
Mechanical	Less damage taken; more ammunition and bombs carried.
Stamina	The odds of surviving when you are shot down.

Rank and decorations. Score earns promotion from 2nd Lieutenant up to Lieutenant Colonel. Five decorations can be won: the Air Medal, the Distinguished Flying Cross for five aerial victories, the Silver Star for an outstanding sortie, the Purple Heart and the Distinguished Unit Citation.

Going down. Being shot down does not always end a career. Depending on his stamina a pilot may bail out and walk home, come back wounded, or be killed. A dead pilot is remembered on the squadron roster, and a replacement you create takes over the campaign where it stands.

Progress is saved after every sortie.

Six days in June

The slice contains two missions of each kind, rising in difficulty. Any of the three mini-games can also be started straight from the menu as **Instant Action**, which does not affect your career.

DATE	MISSION	TYPE	WHERE	OBJECTIVE	DIFFICULTY
June 2nd	Fighter Sweep	Air combat	Pas-de-Calais	Destroy the enemy fighters	
June 4th	The Rail Yard	Bombing	Amiens	Destroy rolling stock and warehouses	
June 6th	D-Day	Ground attack	Road to Caen	Destroy the convoy	
June 10th	Bandits over the Beachhead	Air combat	Sainte-Mere-Eglise	Destroy the enemy fighters	
June 14th	The Bridge	Bombing	River Orne	Destroy the bridge	
June 18th	Airfield Attack	Ground attack	Evreux	Destroy parked aircraft and fuel	

In the second air combat mission Focke-Wulf 190s join the Bf 109s and you fly with two wingmen instead of one. The later bombing and ground attack missions face heavier flak.

Controls and how to run it

ACTION	KEYBOARD	GAMEPAD
Steer	W A S D or arrow keys	Left stick
Guns and bombs	Space	A, right trigger
Throttle (air combat)	Shift to open, Ctrl to close	Shoulder buttons
Pause	Esc	Start
Fullscreen	F11	

- **Air combat:** S pulls the nose up, W pushes it down, A and D roll.
- **Bombing:** steer in all four directions over the scrolling ground.
- **Ground attack:** A and D slide sideways, W dives, S climbs.

The exported build

A Windows build is in the export folder of the project. Keep the two files together; the game will not start without the DLL beside it.

```
export\WW2Wings.exe  
export\ww2wings.dll
```

From source

Requires Windows, Godot 4.7, Visual Studio 2022 with the C++ workload and CMake.

```
git submodule update --init  
build.bat  
run.bat
```

The first build compiles the Godot C++ bindings and takes a few minutes.

Everything is made in code

The game is written in C++ as a Godot 4.7 GDExtension, in roughly 10,900 lines. The Godot project itself is a single scene with a single node. The repository contains no models, textures or audio.

- **Aircraft** are lofted from cross-sections and aerofoil profiles. Paint, camouflage, invasion stripes, national markings and panel lines are drawn by a shader.
- **The landscape** is a shader too: the irregular fields and hedgerows of Normandy, rivers, roads, railways and runways.
- **Sound** is synthesised when the game starts: engines, guns, explosions, the falling-bomb whistle and the music.
- **Lighting** uses Godot's Forward+ renderer with a procedural sky, shadows, ambient occlusion, fog and bloom.



The model viewer: every aircraft, vehicle and building is generated at start-up.

STATUS OF THE VERTICAL SLICE

- The full loop plays from menu to ending, and has been run unattended from start to finish.
- Player controls and the pause menu were checked with simulated key presses.
- Difficulty has only been tuned against a simple autopilot. **It has had no structured play-testing.**
- Sound was checked numerically, not by ear.